Patrick Thangarajah

FINITE ELEMENT ANALYSIS | ENGINEERING MECHANICS | LS-DYNA | MODEL VALIDATION

Calgary, AB | 403-585-1903 | ptemmanual@gmail.com | www.linkedin.com/in/patrickthangarajah/ | TN eligible

Summary

Completed an MASc in Mechanical and Mechatronics Engineering at the University of Waterloo in 2026, achieving a 90% graduate coursework average. Developed analytical, statistical modelling and communication skills through simulation studies, physical test validation, machine learning and sponsor presentations, including to Transport Canada, across medical devices, vehicle safety and manufacturing.

Skills

Finite Element Analysis	LS-DYNA, LS-PrePost, nonlinear FEA, material modeling, contact, boundary conditions
Model Validation and Testing	FE model validation, physical test correlation, ISO 18571, experimental data analysis
Programming and Engineering Tools	MATLAB, Python, C/C++, LS-PrePost scripting, Microsoft Excel, SolidWorks
Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*

- Research in impact mechanics, finite element analysis, and model validation
- Graduate coursework average: 90%

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option

Work/Research Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*

- Validated a THOR-5F ATD FE model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Designed and analyzed a 104-simulation FE parametric study of seat belt anchor locations using JMP DOE, least-squares regression, and ANOVA; identified a rearward and inboard position with 16% lower peak chest deflection than the study average
- Presented research findings through a master's thesis, conference paper, technical reports and six sponsor presentations, including to Transport Canada

Teaching Assistant - MTE 120: Circuits — *Waterloo, ON, Canada*
University of Waterloo — *May 2024 - Aug 2024*

- Supported laboratory instruction for a 100+ student circuits course, troubleshooting circuit analysis, measurement, and hardware issues and providing feedback on laboratory work and exams

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*

- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*

- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Jan 2024 - Apr 2024*

- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

Simulation of Cervical Fractures in American Football (ME 720 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Dec 2023*

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.